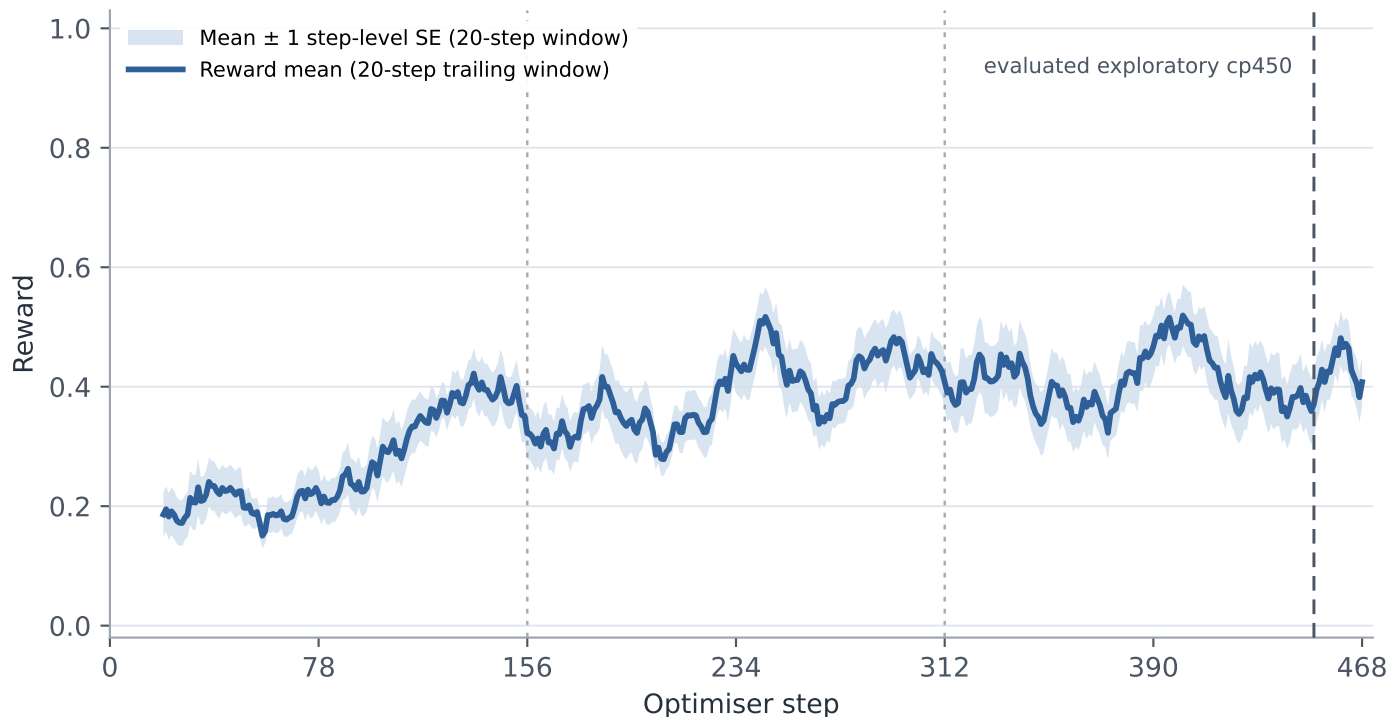


Model chk-3 GRPO: Reward Mean with ± 1 SE

20-step trailing reward summaries; eight completions per optimiser step

Epoch reward mean $\pm$ step-level SE: E1 0.269 ± 0.015 | E2 0.391 ± 0.017 | E3 0.413 ± 0.019



Note: SE = SD of 20 optimiser-step means / $\sqrt{20}$; each mean averages 8 completions.

Descriptive only: no adjustment for serial, overlapping-window, or between-run dependence; not a confidence interval.